

Foundation (Jalapeno)

- Q1.** If $f(x) = x + 2$ and $g(x) = 2x - 1$, what is $f(g(2))$?
(A) 5
(B) 7
(C) 3
(D) 9
(E) 11
- Q2.** Given $f(x) = x^2$ and $g(x) = x - 3$, find $f(g(4))$.
(A) 3
(B) 7
(C) 1
(D) 9
(E) 5
- Q3.** The function $h(x) = f(g(x))$ is defined when $g(x)$ is defined and $g(x) \geq -2$. If $g(x) = 1 - \frac{1}{x}$, what is the domain of $h(x)$?
(A) all real numbers except $x = 0$ and $x = -2$
(B) all real numbers except $x = 0$
(C) all real numbers
(D) $x \geq -2$ and $x \neq 0$
(E) $x < -2$ or $x > 0$
- Q4.** The stock price of a company is modeled by the function $s(x) = 2x + 5$, where x is the number of years since the company's IPO. If $t(x) = x^2$ represents the tax on the stock, what is $t(s(3))$?
(A) 11
(B) 81
(C) 121
(D) 169
(E) 81
- Q5.** A savings account earns interest according to the function $i(x) = 0.05x$, where x is the initial deposit. If $d(x) = x + 1000$ represents the total amount in the account after a deposit, what is $i(d(500))$?
(A) 30
(B) 50
(C) 75
(D) 25
(E) 35
- Q6.** The cost $c(x)$ of producing x units of a product is given by $c(x) = 2x + 1000$. The revenue $r(x)$ from selling x units is given by $r(x) = 10x - 0.1x^2$. What is $r(c(100))$?
(A) 1200
(B) 2800
(C) 3000
(D) 2100
(E) 1100
- Q7.** If $f(x) = x^3$ and $g(x) = \sqrt{x}$, what is the domain of $f(g(x))$?
(A) $x \geq 0$
(B) all real numbers
(C) $x > 0$
(D) $x \leq 0$
(E) $x \neq 0$
- Q8.** The number of employees in a company is modeled by $e(x) = x + 50$, where x is the number of years since the company was founded. If $b(x) = 1000 + 50x$ represents the budget for employee salaries, what is $b(e(5))$?
(A) 2000
(B) 3000
(C) 4000
(D) 3500
(E) 4500
- Q9.** A company's profit $p(x)$ is given by $p(x) = 2x - 500$, where x is the number of units sold. The cost $c(x)$ of producing x units is given by $c(x) = x + 200$. Find $p(c(300))$ and interpret the result in the context of the company's profit.
- Q10.** The value $v(x)$ of an investment after x years is given by $v(x) = 1000(1 + 0.05x)$. The tax $t(x)$ on the investment is given by $t(x) = 0.1x$. Find $t(v(10))$ and explain what this represents in the context of the investment.

Open Ended Questions (FRQ)

- Q9.** A company's profit $p(x)$ is given by $p(x) = 2x - 500$, where x is the number of units sold. The cost $c(x)$ of producing x units is given by $c(x) = x + 200$. Find $p(c(300))$ and interpret the result in the context of the company's profit.
- Q10.** The value $v(x)$ of an investment after x years is given by $v(x) = 1000(1 + 0.05x)$. The tax $t(x)$ on the investment is given by $t(x) = 0.1x$. Find $t(v(10))$ and explain what this represents in the context of the investment.

Chef's Tips

- Remind students to check the domain restrictions of composite functions.
- Encourage students to use real-world examples to interpret the results of composite functions.
- Emphasize the importance of understanding the order of operations when evaluating composite functions.